

Classical Image Processing for Brain Tumor Segmentation in MRI

Group 18 — Project 20
ECE501: Digital Image Processing

TABLE I
GROUP MEMBERS

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Abstract—In this project we are using classical image processing methods such as Histogram Equalization, thresholding, Normalization etc. to segment brain tumors in Magnetic Resonance Imaging (MRI). We have implemented the intensity based thresholding and morphological processing in this pipeline to predict a Whole Tumor (WT) mask. After the analytic prediction we compare the parameters of our results with the original ground truth provided in the BraTS2020 training dataset. The results discovers the mean patient-wise Dice score to be 0.665 ± 0.179 and mean precision to be 0.787 while mean recall to be around 0.600. In overall performance the pipeline can detect tumor regions but often misses some part of the tumor. This report presents the results, limitations and possible improvements for the pipeline version 1.

I. KEYWORDS

Index Terms—brain tumor, MRI, thresholding, morphological processing, segmentation, BraTS, Pipeline

II. INTRODUCTION

In this project our main scope is to identify and localize the tumor region in an MRI scan. Using classical image processing methods, helps us not only in just localizing the tumor but also to estimate the extent of the tumor. Due to the predefined scope of the project, we are mainly focused on tumor segmentation, localization and a small amount of morphological analysis including the area (in pixels and mm) of the tumor region, volume, slices, etc. Our project structure is mainly in parts of pipeline and it is not directed towards a machine learning or deep learning model but mainly focused towards improving the results of different kinds of pipeline combinations using fixed intensity thresholds. Our previous studies have explored that thresholding and morphological operations with a fixed threshold may always not be the best case for every scenario [1], [3]. For the testing we are using BraTS 2020 Training Dataset and for the results of this report we have used a total of 20 samples from the dataset.

III. METHODOLOGY

A. Segmentation and Mask Prediction

We are using multi modal MRI data and some constant intensity based rules to identify the possible tumor voxels (3 dimensional volume element). The main segmentation step happens while thresholding. The voxels who pass a certain criteria in thresholding are identified as candidate voxels and using multiple MRI inputs provides us with complementary intensity information for the candidate tumor region. The result of the foretold process leads to a binary mask which is then refined using 3-D morphological processing, which removes the unwanted regions and gives us binary WT (Whole Tumor) mask [1]–[3].

B. Evaluation

For the evaluation, we have also designed a measurement script (range_statistics.py), which takes all the output data (.csv) as input for a certain user assigned range and calculates decision scores based on certain standard measurement units such as IoU (Intersection over Union) and Dice score between the predicted mask and the ground truth already provided in the BraTS2020 Training Dataset. We also calculate the HD95 and ASSD (Average Symmetric Surface Distance) for boundary difference between the predicted mask and the ground truth for more precision.

IV. RESULTS

Table II summarises the WT results for the 20 evaluated cases.

TABLE II
RESULTS FOR 20 BRA TS CASES (WT)

Metric	Mean $\pm$ SD	Median
Dice	0.665 ± 0.179	0.714
IoU	0.521 ± 0.183	–
Precision	0.787 ± 0.179	–
Recall	0.600 ± 0.211	–
HD95 (mm)	28.12 ± 20.71	25.29
ASSD (mm)	6.96 ± 5.03	5.23
Volume similarity	0.816 ± 0.127	–

The pooled Dice score was 0.733. Dice was at least 0.50 for 16 of 20 cases (80%) and at least 0.70 for 11 cases (55%). The highest Dice scores were 0.888 and 0.850, while the lowest were 0.247, 0.342, and 0.405.

Figure 1 shows an example of the MRI after preprocessing, while Fig. 2 shows the corresponding segmentation output produced by the pipeline. (For data "BraTS20_Training_017")

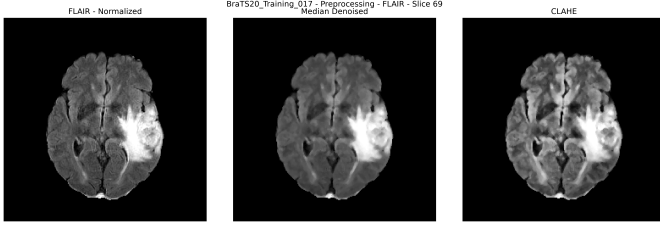


Fig. 1. Example of a preprocessed MRI image.

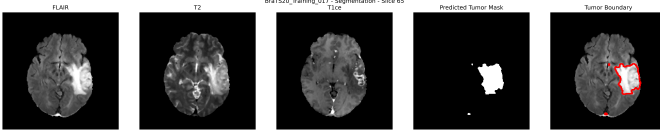


Fig. 2. Example of the predicted tumor segmentation.

V. DISCUSSIONS

In the case of these 20 samples, mean precision (0.787) was higher than mean recall (0.600), which indicates that the predicted masks were generally conservative. Most of the predicted region matched the reference tumor region but parts of the reference tumors were missed. Our analysis classified thirteen cases as under segmentation, three as mixed boundary errors, two as over segmentation and two as excellent overlap. In the histogram below we can observe the Dice and IoU for each patient.

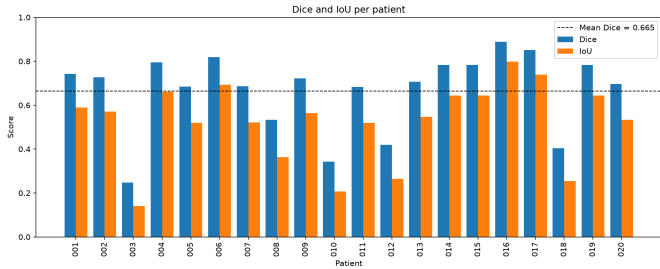


Fig. 3. Dice and IoU for each patient.

The mean signed volume error for us was about 23.3% and 17 cases had underestimated volumes mainly due to the under segmentation and the remaining three had a little over estimated the volumes. The standard deviation of Dice was about 0.179 and the spearman correlation between Dice and tumor size was $\rho = 0.65$, with lower overlap for smaller tumors. These results are limited to 20 cases and the WT region and do not establish clinical suitability. The Figure 4

shows the scatter plot with the predicted vs ground-truth tumor volume accuracy correlation.

$$\text{Dice} = \frac{2 \cdot \text{IoU}}{1 + \text{IoU}}$$

$$\text{IoU} = \frac{\text{Dice}}{2 - \text{Dice}}$$

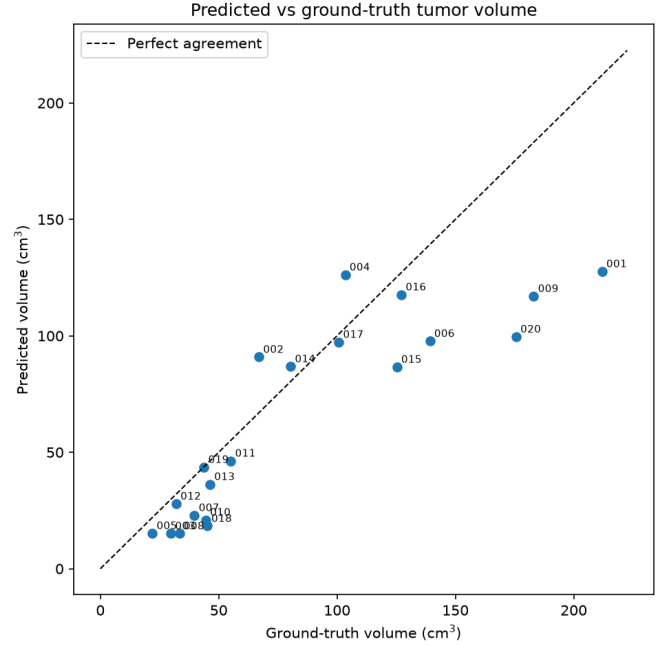


Fig. 4. Predicted and reference tumor volumes.

VI. CONCLUSION

Our classical thresholding and morphology pipeline was evaluated on 20 BraTS training cases and achieved a mean Dice score of 0.665 and a pooled Dice score of 0.733. The main limitation in this version for us was under segmentation along with boundary errors. In our next phase we will try to improve this pipeline or use any other more suitable method for segmentation to optimize the voxel selection. Each possible change in future will be evaluated using the same metrics and cases.

REFERENCES

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